

2024

Time : 3 hours

Full Marks : 70

Pass Marks : 32

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer any **five** questions in which*

Q. No.1 is compulsory.

1. Answer any **four** questions of the following :

$$3\frac{1}{2} \times 4 = 14$$

- (a) What is Fermi's golden rule ?
- (b) What is the difference between degenerate and non-degenerate perturbation ?
- (c) Define scattering amplitude.
- (d) What do you mean by negative energy states ?

- (e) Define density of states of a Quantum Wire.
- (f) Describe, in brief, the shape of C_{60} .
2. Discuss Stark effect using first order quantum mechanical time independent degenerate perturbation theory. 14
 3. Describe Einstein's A and B coefficients as an application of time dependent perturbation theory. 14
 4. Discuss the theory of scattering in laboratory frame of reference. 14
 5. State and derive Klein-Gordon Equation and discuss its merits and demerits. $10+4 = 14$
 6. Set up Dirac's relativistic wave equation and discuss the properties of Dirac's matrices. $8+6 = 14$
 7. Discuss the synthesis of Carbon Nanotubes. Also, describe the electrical and magnetic properties of CNTs. $8+6 = 14$

8. Describe metal nanoclusters. Also, define magic numbers of nano-particles. $8+6 = 14$

9. Write short notes on any **two** of the following :

$7 \times 2 = 14$

(a) Variational methods

(b) Born approximation

(c) Free particle solution of Dirac equation

(d) Preparation of quantum nano structures

